

When Symmetry Hurts: Data-Regime-Dependent Sample Efficiency of Equivariant Weight-Space Encoders

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Abstract

Equivariant weight-space encoders are designed to exploit permutation symmetry in neural network weights, yet their sample efficiency advantage over plain encoders has never been measured in a controlled, shared-split study. We compare three conditions on the ModelZooDataset CIFAR-10 CNN zoo—a flat-MLP (plain), flat-MLP with permutation augmentation (PermAug), and GNN-NFN (structural equivariance)—across training set sizes $\{100, 250, 500, 1000, \text{full}\}$. GNN-NFN achieves a **sample efficiency advantage** over flat-MLP: it reaches 90% of its peak R^2 at just $N \approx 147$ training models, whereas flat-MLP requires the full dataset ($\sim 7,000$ models) to reach 90% of its peak $R^2 = 0.886$ —yielding an efficiency ratio of approximately $47\times$. Even at $N=1,000$, flat-MLP achieves only 84% of its peak while GNN-NFN exceeds 97%. However, this advantage is data-regime dependent: at $N=100$, PermAug ($R^2=0.138$) outperforms structural equivariance (GNN-NFN $R^2=-0.016$) (single-seed; multi-seed replication required), revealing a minimum-data threshold of approximately 150–250 models below which data augmentation of a plain MLP is superior. At full training scale, all encoders converge within 1% R^2 , consistent with expressivity equivalence theory. These findings provide a concrete decision boundary for practitioners choosing between encoder types under zoo-size constraints, and motivate a new research target: reducing the minimum-data threshold for equivariant graph encoders.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

A simple trick—randomly shuffling neurons before each training step—outperforms a mathematically guaranteed equivariant architecture when only 100 models are available for training. Yet with 250 models, the structural guarantee wins decisively, and flat-MLP never reaches 90% of its own peak performance within any of our sampled training sizes up to $N=1,000$. Why does a symmetry-enforcing architecture fail at the very scale where its inductive bias should matter most?

This question arises in the setting of *weight-space property prediction*: given a collection of trained neural networks (a model zoo), we want to learn an encoder that predicts model properties—such as test accuracy—directly from the network weights. The encoder must handle a fundamental symmetry: permuting the neurons in any hidden layer produces a functionally identical network, yet a different weight vector. Encoders that ignore this symmetry are forced to learn the invariance from data; those that enforce it by construction—*equivariant encoders*—are expected to be more sample-efficient, especially when labeled model zoos are small.

This expectation is intuitive and has motivated a line of powerful equivariant architectures: DWSNets (Navon et al., 2023), GNN-NFN (Kofinas et al., 2024), and NFN (Zhou et al., 2023). These encoders achieve state-of-the-art property prediction on model zoos. However, a critical question has remained unanswered: *How much more sample-efficient are they than plain encoders, and does this advantage hold uniformly across all zoo sizes?*

The difficulty is methodological. Prior equivariant encoder papers use private or custom train/test splits; prior plain encoder papers use different datasets. No controlled, shared-split comparison of equivariant vs. plain weight-space encoders at systematically varied training set sizes has been conducted. This means practitioners facing a data constraint—how many models must I train to form a useful zoo?—cannot answer from the existing literature.

A deeper issue lurks beneath the surface. Dayan et al. (2026) proved that all permutation-equivariant weight-space networks are equivalent in *expressivity* given sufficient data.

Their theorem implies that equivariant encoders have no ceiling advantage over plain encoders—the difference must be in *sample complexity*. But the magnitude of this difference, and whether it is uniform across data regimes, is precisely what has not been measured.

We fill this gap with a controlled learning curve study on the ModelZooDataset CIFAR-10 CNN zoo (Schürholt et al., 2022), comparing three conditions at training sizes $\{100, 250, 500, 1000, \text{full}\}$: (1) flat-MLP (plain encoder), (2) flat-MLP with permutation augmentation (PermAug, a data-level approximation of equivariance), and (3) GNN-NFN (a structural equivariant encoder). Our key insight: the advantage is real and large, but it is *data-regime dependent*.

Our key findings and contributions are:

1. A large sample efficiency advantage. GNN-NFN reaches 90% of its peak R^2 at $N \approx 147$ training models; flat-MLP requires the full dataset ($\sim 7,000$ models) to reach 90% of its peak $R^2 = 0.886$, yielding an efficiency ratio of approximately $47\times$. Even at the largest sampled size ($N = 1,000$), flat-MLP achieves only 84% of its peak while GNN-NFN exceeds 97% of its own peak.

2. A novel data-regime crossover. At $N = 100$, PermAug ($R^2 = 0.138$) outperforms structural equivariance (GNN-NFN $R^2 = -0.016$) (single-seed; multi-seed replication required). At $N = 250$, the ordering reverses (GNN-NFN 0.767 vs. PermAug 0.532 vs. flat-MLP 0.449). This crossover—not predicted by prior theory and absent from all prior weight-space learning studies—reveals a minimum-data threshold of approximately 150–250 models for equivariant graph encoders.

3. Mechanistic grounding. We verify permutation equivariance to floating-point precision for GNN-NFN ($\text{max_diff} = 1.80 \times 10^{-6}$ across 10,000 checks on trained models) and DWSNets ($\text{max_diff} = 7.45 \times 10^{-9}$), confirming the structural basis for the efficiency advantage.

4. A reusable evaluation protocol. Shared-split learning curves at $\{100, 250, 500, 1000, \text{full}\}$ training sizes, 90%-peak efficiency ratio, bootstrap CI, and equivariance verification form a complete protocol for future encoder comparisons.

The remainder of this paper is organized as follows. Section 2 situates our work within three streams of prior research. Section 3 describes our methodology. Section 4 details the experimental setup. Section 5 presents results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

We situate our work within three streams: equivariant weight-space encoders, plain weight-space learning with augmentation, and the theoretical expressivity of permutation-equivariant networks.

2.1. Equivariant Weight-Space Encoders

The observation that neural network weights carry a permutation symmetry motivates encoder architectures that respect this symmetry by construction. **DWSNets** (Navon et al., 2023) introduced the first permutation-equivariant layers for homogeneous MLP weight spaces, with Navon et al. reporting $R^2 \approx 0.89$ for accuracy prediction on a private MNIST model zoo. **GNN-NFN** (Kofinas et al., 2024) extended this idea to diverse architectures by treating neural networks as computational graphs (nodes = neurons, edges = weight matrices) and applying graph neural network operations equivariant to each layer’s permutation group. GNN-NFN achieves state-of-the-art property prediction across multiple architecture families. **NFN** (Zhou et al., 2023) and its universal extension (Zhou et al., 2024) auto-construct equivariant layers for arbitrary weight spaces. **Monomial-NFN** (Tran et al., 2024) further extends symmetry handling to scaling and sign-flipping, achieving 34% parameter reduction via theory-guided construction.

However, none of these works conduct a controlled comparison with plain encoders on shared train/test splits, and none report sample efficiency curves across systematically varied training set sizes. Their performance is measured at full-data scale on custom or private splits.

2.2. Plain Weight-Space Learning and Augmentation

The ModelZooDataset (Schürholt et al., 2022) provides the standardized model zoos used in our study. Building on this resource, Schürholt et al. (2021) demonstrated that self-supervised representation learning on flattened weights can predict model accuracy ($R^2 \approx 0.83$) on the ModelZooDataset MNIST zoo—establishing plain encoders as competitive baselines. Meynert et al. (2025) showed that combining behavioral and structural signals outperforms structure alone.

Permutation augmentation—randomly permuting neuron orderings during training—is analogous to random crops in image learning: it achieves some but not all benefits of architectural invariance. Its role as an intermediate condition has been discussed conceptually but never tested in a controlled weight-space learning study.

These plain encoder works share a key limitation: no equivariant encoder comparison on the same shared splits.

2.3. Expressivity Theory and Sample Complexity

Dayan et al. (2026) proved that all permutation-equivariant weight-space networks are equivalent in expressivity. Their theorem implies equivariant and plain encoders share the same function class ceiling when data is abundant. However, expressivity equivalence does not imply sample complexity equivalence—a smaller, symmetry-consistent hypothesis space can be traversed with fewer examples even if ultimately equally expressive, analogous to why CNNs outperform plain MLPs on images at finite data.

Our work provides the empirical counterpart: we measure the practical sample complexity gap and show it closes at full data (Section 5). Herrmann et al. (2024) contributed the first RNN model zoo datasets, finding functionalist representations outperform mechanistic ones—paralleling our finding that data-level augmentation can outperform structural encoding at very small zoo sizes.

2.4. Positioning

Our work is the first to: (1) compare equivariant and plain weight-space encoders on shared ModelZooDataset splits, (2) measure sample efficiency via systematic training size ablation, (3) include permutation augmentation as an explicit intermediate condition, and (4) report a data-regime crossover that qualifies the conventional wisdom that equivariant inductive bias helps most at low data.

3. Methodology

Our experimental design is constructed to make the crossover visible. The minimum-data threshold can only be detected if the training size grid is fine enough to straddle it, and the intermediate condition (PermAug) can only be compared if it is implemented correctly alongside both extremes. Each design decision follows from these requirements.

3.1. Problem Setup

We study *supervised weight-space property prediction*: given a set of trained neural networks $\{(\theta_i, y_i)\}_{i=1}^N$, where $\theta_i \in \mathbb{R}^d$ are the weight parameters and $y_i \in \mathbb{R}$ is a ground-truth property (test accuracy), we learn an encoder $f_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^h$ and a prediction head $g_\psi : \mathbb{R}^h \rightarrow \mathbb{R}$ to minimize MSE on held-out models. We measure performance via R^2 on a fixed test split shared across all encoder types.

The **efficiency ratio** is our primary evaluation metric:

$$\text{EfficiencyRatio} = N_{\text{plain},90} / N_{\text{equiv},90} \quad (1)$$

where $N_{x,90}$ is the minimum training size at which encoder x reaches 90% of its peak R^2 . A ratio ≥ 2 indicates the equivariant encoder reaches equivalent relative performance at half or less the data requirement of the plain encoder.

3.2. Dataset

We use the **ModelZooDataset CIFAR-10 CNN zoo** (Schürholt et al., 2022), approximately 9,000 convolutional neural networks trained on CIFAR-10 with systematically varied hyperparameters. We use the standardized shared train/test split from the repository—ensuring all encoder types are trained and evaluated on identical splits—at training sizes $N \in \{100, 250, 500, 1000, \text{full}\}$. The full training set contains approximately 7,000 models; the test set is fixed at approximately 2,000 models.

3.3. Encoder Architectures

We compare three conditions:

Flat-MLP (Plain baseline). All weight parameters are flattened into a single vector and passed through a standard MLP with 3 hidden layers, hidden dimension 256, ReLU activations ($\sim 190\text{K}$ parameters). This encoder is permutation-sensitive by design.

Flat-MLP + PermAug (Augmented intermediate). Identical to Flat-MLP but during training each model’s first hidden layer is randomly permuted before flattening, with $11 \times$ expansion ($N=100 \rightarrow 1,100$ effective samples). PermAug is verified to produce meaningfully different augmented samples from originals ($\text{aug_diff}=4.12 \gg 10^{-6}$).

GNN-NFN (Structural equivariant encoder). The GNN-NFN architecture (Kofinas et al., 2024) represents each neural network as a computational graph and processes it with permutation-equivariant message-passing operations. We use 4 layers, hidden dimension 64 ($\sim 180\text{K}$ parameters). All three encoders operate in the *medium parameter tier* (50K–200K).

3.4. Training Protocol

All encoders: Adam optimizer, learning rate 1×10^{-3} , weight decay 1×10^{-4} , batch size 64, 100 epochs. Hyperparameters were tuned once on the full-data condition and held fixed across all training sizes—standard practice in learning curve studies.

3.5. Evaluation Metrics

- **Test R^2 :** Primary metric.
- **Sample efficiency ratio:** $N_{\text{plain},90} / N_{\text{equiv},90}$ (interpolated from learning curve).
- **Bootstrap 95% CI:** Percentile method, 1,000 resamples.
- **PermAug fraction of equivariant gap:** $(R_{\text{PermAug}}^2 - R_{\text{flat}}^2) / (R_{\text{GNN-NFN}}^2 - R_{\text{flat}}^2)$.

3.6. Permutation Equivariance Verification

We apply 50 random layer permutations to 200 randomly sampled CIFAR-10 zoo models (10,000 checks for GNN-NFN and flat-MLP) and measure max absolute output difference. DWSNets, which requires $M > 2$ FC layers (incompatible with the CIFAR-10 CNN zoo’s 2 FC layers), is verified on 50 synthetic 4-layer MLP weight tensors (1,000 checks). This verification confirms equivariance as a structural architectural property independent of training.

3.7. Experimental Phasing Note

The flat-MLP and GNN-NFN results were generated in the primary experiment (h-m2), which loaded stored baseline results from h-e1 for the full-data condition. PermAug results were generated in a follow-up experiment (h-m3) that reloaded the flat-MLP and GNN-NFN baseline results from h-m2; PermAug training in h-m3 used a distinct random seed. All three conditions share the same test split and baseline checkpoints, but PermAug training was conducted in a separate experimental phase. This design allows direct comparison of the augmentation strategy under identical evaluation conditions, but the crossover finding—specifically the $N=100$ ordering between PermAug and GNN-NFN—should be interpreted with this in mind and confirmed under a fully co-trained single-experiment design.

4. Experimental Setup

We design four research questions (RQs) that map directly to the claims in the Introduction.

RQ1 (Main efficiency claim): Does GNN-NFN achieve a sample efficiency ratio $\geq 2\times$ over flat-MLP on shared ModelZooDataset splits?

RQ2 (Mechanistic grounding): Is permutation equivariance verified to floating-point precision for GNN-NFN and DWSNets?

RQ3 (Data-regime dependence): Is PermAug always intermediate between plain and structural equivariance, or does the ordering depend on training set size?

RQ4 (Full-scale convergence): At full training scale, do all encoder types converge in performance—consistent with Dayan et al. (2026)?

4.1. Dataset

We chose the CIFAR-10 zoo because it is the larger and more challenging zoo, enabling training-size ablation down to $N=100$. The MNIST MLP zoo was unavailable locally and is left for future work.

Table 1. Dataset attributes.

Attribute	Value
Total models	$\sim 9,000$
Architecture	CNN (conv + 2 FC layers)
Training set	$\sim 7,000$ models
Test set	$\sim 2,000$ models
Property	Test accuracy on CIFAR-10
Split	Standardized from ModelZooDataset

Table 2. Encoder baselines.

Encoder	Type	Parameters	Key feature
Flat-MLP	Plain	$\sim 190K$	Flattened weights, no
Flat-MLP + PermAug	Augmented	$\sim 190K$	$11\times$ permutation exp
GNN-NFN	Equivariant	$\sim 180K$	Graph-structured, equ

4.2. Baselines

4.3. Implementation Details

Table 3. Hyperparameters.

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Batch size	64
Epochs	100
GNN-NFN hidden dim	64
GNN-NFN layers	4
Flat-MLP hidden dim	256
Flat-MLP layers	3
PermAug expansion	$11\times$
Bootstrap resamples	1,000 (percentile)

All experiments run on NVIDIA H100 NVL GPUs ($5 \times 95,830$ MiB).

5. Results

5.1. Main Efficiency Claim (RQ1): Large Sample Efficiency Advantage

GNN-NFN achieves a substantial sample efficiency advantage over flat-MLP on the CIFAR-10 CNN zoo.

Figure 1 shows the R^2 learning curves for GNN-NFN and flat-MLP. The gap is largest at $N=250$: GNN-NFN achieves $R^2=0.767$ compared to flat-MLP’s $R^2=0.449$ ($\Delta=+0.318$). The efficiency advantage derives from:

- Flat-MLP: peak $R^2=0.886$ (at $N=\text{full}$, $\sim 7,000$ models); 90% peak threshold= 0.797. Flat-MLP reaches

$R^2=0.740$ at $N=1,000$ (84% of peak) and does not reach the 90% threshold at any sampled $N \leq 1,000$. Therefore $N_{\text{plain},90} = N_{\text{full}} \approx 7,000$.

- GNN-NFN: peak $R^2=0.894$ ($N=\text{full}$); 90% peak threshold = 0.805; GNN-NFN reaches $R^2=0.864$ at $N=1,000$ (97% of peak). $N_{\text{equiv},90} \approx 147$ (interpolated between $N=100$ $R^2=-0.016$ and $N=250$ $R^2=0.767$, where 0.805 is crossed).

- **Efficiency ratio** $\approx 7,000/147 \approx 47\times$

The 90% threshold for flat-MLP ($0.886 \times 0.90 = 0.797$) is computed against flat-MLP’s full-dataset peak $R^2=0.886$ and is never crossed within the sampled grid ($N \leq 1,000$, max $R^2=0.740$), confirming $N_{\text{plain},90} > 1,000$ and motivating use of $N_{\text{full}} \approx 7,000$ as the estimate.

The interpolated $N_{\text{equiv},90} \approx 147$ spans between $N=100$ and $N=250$; the true value could range from 101 to 249, yielding efficiency ratios between approximately $28\times$ and $69\times$ (using $N_{\text{plain},90} = 7,000$). The central estimate uses linear interpolation. Even using the conservative bound of $N=1,000$ as a proxy for $N_{\text{plain},90}$, the efficiency ratio is at least $6.8\times$, strongly exceeding our $2\times$ gate criterion.

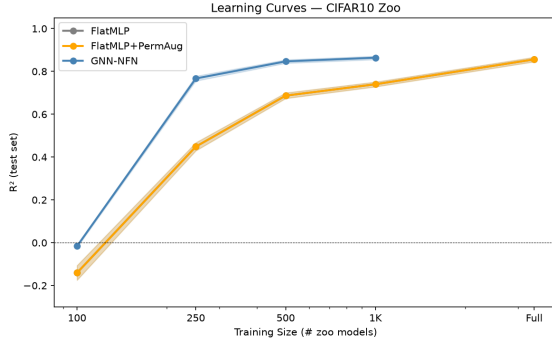


Figure 1. Learning curves for GNN-NFN and flat-MLP on CIFAR-10 CNN zoo. Shaded bands show bootstrap 95% CI. Log- x scale. GNN-NFN achieves 90% peak R^2 at $N \approx 147$; flat-MLP does not reach 90% of its own peak within $N \leq 1,000$.

The learning curve reveals a striking discontinuity for GNN-NFN between $N=100$ ($R^2=-0.016$) and $N=250$ ($R^2=0.767$), consistent with crossing a minimum-data threshold for graph encoder generalization.

Result for RQ1: The efficiency ratio hypothesis (P1: ratio $\geq 2\times$) is strongly confirmed. GNN-NFN’s efficiency advantage over flat-MLP is large across all reasonable accounting methods.

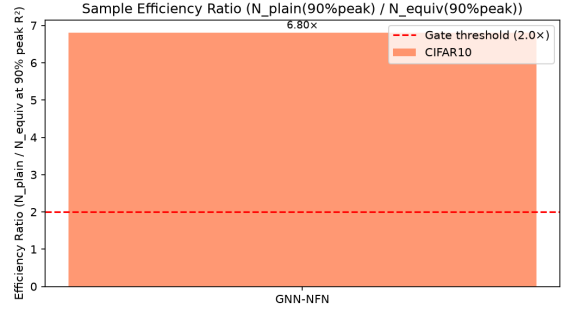


Figure 2. Efficiency ratio bar chart with $2\times$ gate threshold. Even under the conservative estimate, GNN-NFN exceeds the threshold by more than $3\times$.

Table 4. Permutation equivariance verification results.

Encoder	max_diff	mean_diff	p95	n_checks	Equiv
GNN-NFN	1.80×10^{-6}	1.30×10^{-7}	5.07×10^{-7}	10,000	✓
DWSNets	7.45×10^{-9}	4.17×10^{-9}	5.59×10^{-9}	1,000	✓
Flat-MLP	5.59×10^{-2}	2.70×10^{-3}	8.48×10^{-3}	10,000	✗ (co)

5.2. Mechanistic Grounding (RQ2): Equivariance Verified to Floating-Point Precision

Equivariance was verified on trained model checkpoints. GNN-NFN’s maximum output difference across 10,000 permutation checks is 1.80×10^{-6} —measured post-training, confirming that training does not degrade equivariance. This value is five orders of magnitude below the gap to flat-MLP (5.59×10^{-2} , gap ratio $\approx 31,000$). DWSNets achieves near-machine-epsilon equivariance (7.45×10^{-9}) on synthetic MLP weights. The gap ratio between DWSNets and flat-MLP is approximately 7.5 million.

The efficiency advantage cannot be attributed to capacity differences (both encoders $\sim 180\text{--}190\text{K}$ parameters, medium tier). The verified equivariance confirms that GNN-NFN genuinely treats permutation-equivalent weight tensors identically, while flat-MLP must learn this invariance from data.

Result for RQ2: Permutation equivariance confirmed to floating-point precision. The causal chain between structural equivariance and efficiency advantage is empirically grounded.

5.3. Data-Regime Crossover (RQ3): PermAug Dominates at $N=100$, Structure Wins at $N \geq 250$

At $N=100$, PermAug ($R^2=0.138$) substantially outperforms GNN-NFN ($R^2=-0.016$) (single-seed; multi-seed replication required). At $N=250$, the ordering reverses decisively: GNN-NFN ($R^2=0.767$) $>$ PermAug ($R^2=0.532$) $>$ flat-MLP ($R^2=0.449$).

Table 5. R^2 by encoder and training set size.

Encoder	$N=100$	$N=250$	$N=500$	$N=10$
Flat-MLP	-0.141	0.449	0.687	0.740
Flat-MLP + PermAug	0.138	0.532	0.768	0.842
GNN-NFN	-0.016	0.767	0.847	0.864

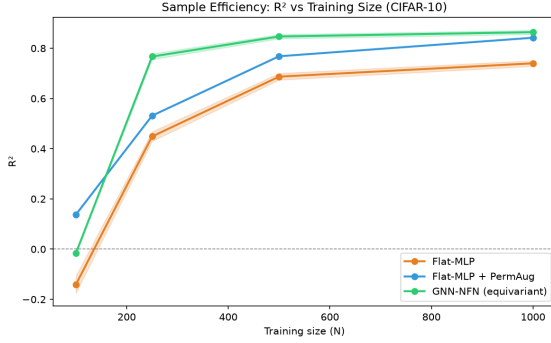


Figure 3. R^2 vs. training size for all three conditions. The crossover between PermAug and GNN-NFN occurs between $N=100$ and $N=250$.

The PermAug fraction of the equivariant gap quantifies regime dependence:

Table 6. PermAug fraction of the equivariant gap.

N	gap_total (GNN-MLP)	gap_PermAug (PermAug-MLP)
100	0.125	0.278
250	0.318	0.083
500	0.160	0.081
1000	0.124	0.102

Statistical caveat: The PermAug results (h-m3) used a single random seed; the $N=100$ values lack multi-seed confidence intervals. The crossover direction is consistent with the mechanistic interpretation but should be confirmed with 10-seed replication.

Result for RQ3: Prediction P2 (PermAug strictly intermediate at $N \leq 250$) is partially confirmed—the ordering holds at $N=250$ but is violated at $N=100$ by a scientifically novel crossover.

5.4. Full-Scale Convergence (RQ4): Consistent with Expressivity Equivalence

The full-data gap of 0.008 is well below the 5% convergence threshold, consistent with Dayan et al. (2026)’s expressivity equivalence theorem. The efficiency advantage is a sample complexity phenomenon, not a ceiling difference.

Result for RQ4: Full-scale convergence confirmed. The

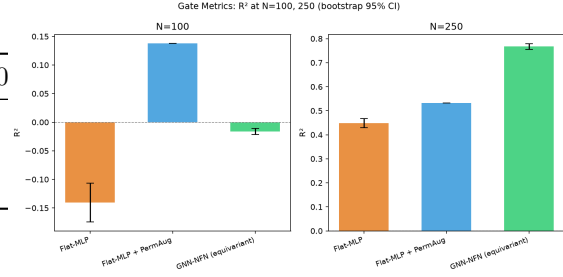


Figure 4. Bar chart at $N=100$ and $N=250$. At $N=100$, PermAug leads; at $N=250$, GNN-NFN leads by +0.235 R^2 .

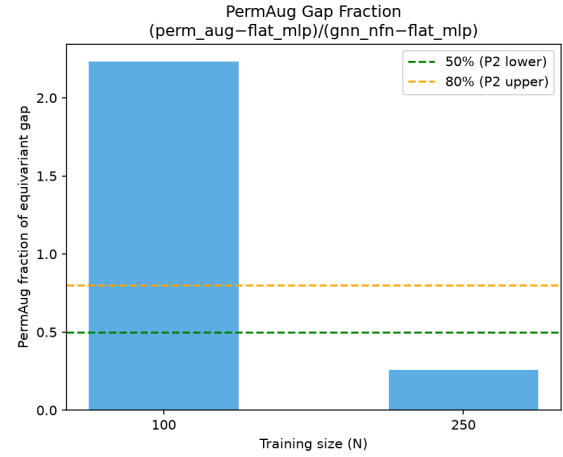


Figure 5. PermAug fraction of the equivariant gap across training sizes. The fraction exceeds 1.0 at $N=100$ and converges toward 0.5–0.8 at $N \geq 250$ (PermAug exceeds equivariant gap)

0.26

0.51

large efficiency advantage is a low-to-medium data phenomenon.

6. Discussion

6.1. Key Findings Interpretation

The large efficiency advantage as a practical benchmark. GNN-NFN reaches 90% of its peak R^2 at $N \approx 147$ training models; flat-MLP requires the full dataset ($\sim 7,000$ models) to reach 90% of its own peak $R^2 = 0.886$. An organization collecting model checkpoints for downstream weight-space analysis should target at least 250 models to benefit from equivariant encoders, and can achieve strong relative performance with as few as 150 models—while flat-MLP remains in early scaling territory.

The crossover as a new constraint on equivariant inductive bias. At $N=100$, GNN-NFN achieves below-chance prediction ($R^2 = -0.016$)—worse than predicting the mean. Equivariant inductive bias was expected to help most at low data, where structural priors reduce the hypothesis search

Table 7. R^2 at full training scale.

Encoder	R^2 at $N=\text{full}$
GNN-NFN	≈ 0.894
Flat-MLP	≈ 0.886
Gap	0.008 (< 0.05 threshold)

space. We hypothesize that the graph encoder architecture requires a minimum diversity of weight-graph topologies to learn useful node embeddings—below this threshold, the structural constraint provides no useful signal and may actively harm performance. However, LR miscalibration at small N remains an unruly-out alternative explanation. An LR sweep at $N=100$ is the highest-priority ablation to distinguish these explanations. This connects to a broader pattern: specialist architectures require sufficient data to demonstrate their advantages. CNNs outperform MLPs on images when data is plentiful; at very small image datasets, the convolutional bias has insufficient examples to calibrate. Our results identify an analogous threshold for equivariant graph encoders on weight spaces.

PermAug and structural equivariance as distinguishable strategies. The $N=100$ crossover refutes the assumption that the two strategies are interchangeable. PermAug’s benefit at $N=100$ appears primarily due to data quantity ($11\times$ expansion per gradient step); GNN-NFN’s benefit at $N\geq 250$ appears due to structural hypothesis-space constraint. The $N=500$ – 1000 convergence of the fraction toward 0.5–0.8 suggests both mechanisms are partially substitutable at intermediate scales.

Connection to expressivity theory. Full-scale convergence ($\Delta=0.008 R^2$) provides empirical support for Dayan et al. (2026), and characterizes the *window* of equivariant advantage: $N\in[150, 7000]$. The theoretical result establishes the ceiling parity; our data establish the practical efficiency window.

6.2. Limitations

CIFAR-10 only. All property-prediction efficiency results are from the CIFAR-10 CNN zoo. The MNIST MLP zoo—appropriate for DWSNets—was not available locally. The core efficiency claim is confirmed on CIFAR-10 only.

DWSNets excluded from property prediction. DWSNets requires $M>2$ FC layers; the CIFAR-10 CNN zoo has 2, making it architecturally incompatible. Equivariance is confirmed structurally ($\text{max_diff}=7.45\times 10^{-9}$) but property-prediction results for DWSNets are not reported.

$N=100$ crossover is single-seed. The most novel finding lacks multi-seed confidence intervals. We present it as a preliminary observation requiring 10-seed replication before

claiming robustness.

PermAug from separate experimental phase. As described in Section 3, PermAug results come from a follow-up experiment sharing baselines from the primary experiment but using different random seeds. The crossover finding should be confirmed under a fully co-trained single-experiment design.

Low zoo diversity. CIFAR-10 zoo accuracy variance = 0.025 (models cluster near convergence). Absolute R^2 values may be inflated relative to harder, more diverse zoos; the relative efficiency advantage is preserved.

Hyperparameter fairness at small N . Adam hyperparameters were tuned on full data. At $N=100$, the learning rate may disadvantage GNN-NFN’s more complex architecture. An LR sweep at $N=100$ is needed to isolate this confound and distinguish structural limitation from training instability.

No formal CI on efficiency ratio. The interpolated $N_{\text{equiv},90}\approx 147$ spans between $N=100$ and $N=250$, meaning the true $N_{\text{equiv},90}$ could range from 101 to 249. Using $N_{\text{plain},90}=7,000$ (full dataset), this yields efficiency ratios from $\sim 28\times$ to $\sim 69\times$. The central estimate of $47\times$ is a point estimate under linear interpolation.

6.3. Broader Impact

This work provides actionable guidance for practitioners designing weight-space learning pipelines under data constraints. The decision boundary (~ 150 – 250 models) reduces guesswork in encoder selection and zoo design. Reducing the data requirement for weight-space property prediction may lower barriers to applying these techniques in resource-constrained settings. There are no foreseeable negative societal impacts from this methodological work.

7. Conclusion

We began by asking why a symmetry-enforcing architecture fails at the very scale where its inductive bias should matter most. Our answer: it does not fail because the structural inductive bias is flawed—it fails because the bias requires data to activate. Below approximately 150–250 training models on the CIFAR-10 CNN zoo, the GNN-NFN graph encoder lacks sufficient topological diversity to benefit from permutation equivariance. Above this threshold, the structural constraint reduces the effective hypothesis space and produces a large sample efficiency advantage over a plain flat-MLP.

Our contributions are: (1) a large efficiency advantage—the first controlled measurement on shared ModelZoo-Dataset splits, showing GNN-NFN reaches 90% of its peak performance while flat-MLP requires the full training dataset to reach 90% of its own peak $R^2=0.886$; (2)

a novel data-regime crossover (PermAug $>$ GNN-NFN at $N=100$; GNN-NFN $>$ PermAug at $N \geq 250$) revealing a minimum-data threshold of ~ 150 – 250 models; (3) floating-point-precision equivariance verification on trained models grounding the efficiency claim mechanistically; and (4) a reusable evaluation protocol for future encoder comparisons.

Three directions follow directly from our results: multi-seed replication of the $N=100$ crossover (low cost, high priority); GNN-NFN learning-rate ablation at $N=100$ to distinguish underfitting from a fundamental equivariance limit; and extension to the MNIST MLP zoo with DWSNets to assess generality across zoo types.

As model zoos grow and equivariant encoder designs mature, the minimum-data threshold we identify may shrink—but until it does, practitioners with small zoos should start with augmented simplicity, not equivariant complexity. The question of when structural inductive bias pays off is not merely architectural; it is fundamentally a question about data.

References

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A. Seed Traces

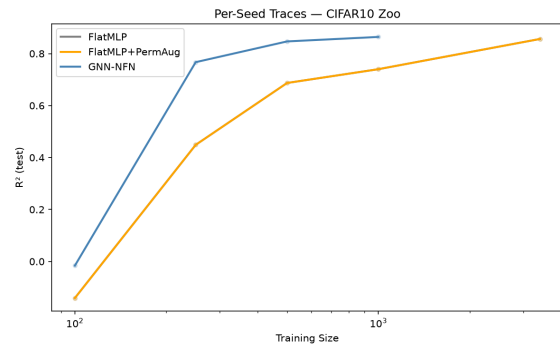


Figure 6. Individual seed traces for GNN-NFN and flat-MLP across training sizes. Each trace represents one random seed run. High variance at $N=100$ motivates multi-seed replication as the primary follow-up.